

Mathematical and Computational Methods

Exercise Sheet 2: Complex numbers I: algebra and geometry

Tutorial route (►, in this order, about 80 minutes): 1.3, 2.1(a,c), 2.3, 2.4, 2.5(a–c), 2.7, 2.10. Physics groups may replace 2.4(b) with 2.6. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★ ★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 (★ drill — powers of i). Simplify the powers of the imaginary unit

$$i^3, \quad i^4, \quad i^{15}, \quad i^{2026}, \quad i^{-1}, \quad i^{-7}.$$

Exercise 1.2 (★ drill — Cartesian arithmetic). With $z = 3 + 2i$ and $w = 1 - 5i$, find $z + w$, $z - w$, zw and z^2 .

Exercise 1.3 (► ★ drill — parts, conjugate and modulus). For $z = 12 - 5i$ find: (a) $\operatorname{Re} z$ and $\operatorname{Im} z$; (b) $\bar{z}$; (c) $|z|$ and $z\bar{z}$; (d) $\operatorname{Im}(iz)$.

Exercise 1.4 (★ drill — division). Write $\frac{4 + 3i}{1 + 2i}$ in Cartesian form $a + bi$.

Exercise 1.5 (★ drill — quadratics over $\mathbb{C}$). Solve $x^2 + 1 = 0$ and $x^2 + 25 = 0$ over $\mathbb{C}$.

Exercise 1.6 (★ drill — an Argand sketch). On a single Argand diagram, plot the points $z_1 = 3 + i$, $z_2 = -2 + 2i$ and $z_3 = -1 - 3i$, together with the sum $z_1 + z_2$. Label each point.

Exercise 1.7 (★ drill — polar and back). Convert to polar form $r(\cos \theta + i \sin \theta)$ with principal argument: (a) $1 + i$; (b) $2i$; (c) -3 . (d) Convert $z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ to Cartesian form.

Exercise 1.8 (★ drill — the polar rules). Let $z = 3(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ and $w = 2(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12})$. Using the polar rules, find zw and z/w in Cartesian form.

2 Tutorial problems

Exercise 2.1 (► ★★ core — the conjugate and modulus rules). Let $z = 2 + 3i$ and $w = 1 - 4i$ (as in the notes).

- Compute zw and z/w in Cartesian form.
- Verify the conjugate rule $\overline{zw} = \bar{z}\bar{w}$.
- Verify the modulus rule $|zw| = |z| |w|$.

Exercise 2.2 (★★ core — two general proofs (maths)). (Maths stream — general proofs.) Let $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$.

- Prove the conjugate-of-a-product rule $\overline{zw} = \bar{z}\bar{w}$ directly from the definition $\overline{a + bi} = a - bi$.

(b) Using (a) and $|u|^2 = u\bar{u}$, prove that $|zw| = |z| |w|$.

Exercise 2.3 (► ★★core — *conjugate pairs*). Solve over $\mathbb{C}$, and confirm that the roots form a conjugate pair: (a) $x^2 + 2x + 10 = 0$, by completing the square; (b) $2x^2 + x + 1 = 0$, by the quadratic formula.

Exercise 2.4 (► ★★core — *plotting and distance (data science)*). (a) Let $z = 3 + 2i$ and $w = -1 + 2i$. Plot z , w , $\bar{z}$, $z + w$, $z - w$ on an Argand diagram, and compute the distance $|z - w|$ between the points z and w .

(b) (*Data science — distance between records.*) Two data points in the plane are encoded as $p = 2 + 5i$ and $q = 6 + 2i$. Compute the Euclidean distance $|p - q|$ between them and the midpoint $\frac{1}{2}(p + q)$.

Exercise 2.5 (► ★★core — *principal argument*). Convert each number to polar form, giving the principal argument $\text{Arg } z \in (-\pi, \pi]$. Sketch each to fix the quadrant.

(a) $1 + i$; (b) $-1 + i$; (c) $-2 - 2\sqrt{3}i$; (d) $\sqrt{3} - i$; (e) -5 .

Exercise 2.6 (★★core — *AC impedance (physics)*). (*Physics — AC circuits.*) A series circuit has complex impedance $Z = 4 + 3i$ (in ohms). It is driven by a voltage $V = 10$ (volts, taken as a real reference).

(a) Find $|Z|$ and the phase angle $\text{Arg } Z$.

(b) Compute the current $I = V/Z$ in Cartesian form and its magnitude $|I|$.

Exercise 2.7 (► ★★core — *polar product and quotient*). Let $z = \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$ and $w = 2(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2})$. Find (a) zw and (b) z/w , giving each in polar form with the principal argument and in Cartesian form.

Exercise 2.8 (★★core — *multiplication by i (with phasors)*). (a) For $z = 2 + i$, compute iz and confirm geometrically that multiplying by i is a quarter-turn anticlockwise that preserves length.

(b) Which single complex number rotates a point by $\frac{\pi}{2}$ anticlockwise and doubles its distance from the origin?

(c) Apply a clockwise quarter-turn to $3 + i$ (i.e. multiply by the number of modulus 1 and argument $-\frac{\pi}{2}$).

(d) (*Physics — phasors.*) A voltage phasor $V_1 = 6 + 8i$ is joined by a second phasor V_2 of equal amplitude turned a quarter-turn anticlockwise, i.e. $V_2 = iV_1$. Find V_2 , the resultant $V = V_1 + V_2$ in Cartesian form, and its amplitude $|V|$.

Exercise 2.9 (★★core — *true or false*). (*Conceptual — true or false, with a one-line reason.*)

(a) For every $z \in \mathbb{C}$, $z^2 \geq 0$.

(b) $|z| = |\bar{z}|$ for all z .

(c) If $z + \bar{z} = 0$ then z is purely imaginary or zero.

(d) $\text{Re}(zw) = \text{Re}(z) \text{Re}(w)$ for all z, w .

(e) There is an ordering $<$ on $\mathbb{C}$ compatible with its arithmetic — that is, one respecting $+$ and $\times$ (if $x > 0$ and $y > 0$ then $x + y > 0$ and $xy > 0$), so that every nonzero square is positive.

Exercise 2.10 (► ★★challenge — *an equilateral triangle*). (*Rotation/geometry.*) Two vertices of an equilateral triangle lie at 0 and 2 on the real axis. Use multiplication by a complex number of modulus 1 to find the two possible positions of the third vertex.

3 Further practice (home)

Exercise 3.1 (★★core — solve for real x, y). Find the real numbers x, y satisfying $(x+yi)(3-i) = 7+i$.

Exercise 3.2 (★★core — division). Simplify to Cartesian form: (a) $\frac{1+2i}{3-4i}$; (b) $\frac{i}{1+i}$.

Exercise 3.3 (★★core — a real cubic from one root). Given that $1+2i$ is a root of $x^3 - 4x^2 + 9x - 10 = 0$ (real coefficients), find the other two roots.

Exercise 3.4 (★★core — loci and a region in the plane). Describe geometrically, and sketch, the set of points $z \in \mathbb{C}$ satisfying

$$(a) |z| = 2; \quad (b) |z - 1| = 3; \quad (c) |z - i| = |z + i|; \quad (d) |z - 1| < 2.$$

For (d), shade the region and show clearly whether its boundary is included.

Exercise 3.5 (★★core — one product, two ways). Compute the product $(1+i)(\sqrt{3}+i)$ in two ways — by Cartesian expansion and by the polar rule — and hence write down its modulus and argument.

Exercise 3.6 (★★core — modulus and argument round-up). Suppose $|z| = 2$, $\text{Arg } z = \frac{\pi}{3}$, $|w| = 3$, $\text{Arg } w = \frac{\pi}{4}$. Find the modulus and argument of (a) zw , (b) z/w , (c) z^2 and (d) $\bar{z}$; (e) convert z and z^2 to Cartesian form.

Exercise 3.7 (★★★challenge — the parallelogram law). Prove the parallelogram law: for all $z, w \in \mathbb{C}$,

$$|z+w|^2 + |z-w|^2 = 2|z|^2 + 2|w|^2.$$

Exercise 3.8 (★★★challenge — a square root in Cartesian form). Find all $z = x + yi$ (with $x, y \in \mathbb{R}$) such that $z^2 = 3 + 4i$, working directly in Cartesian form.